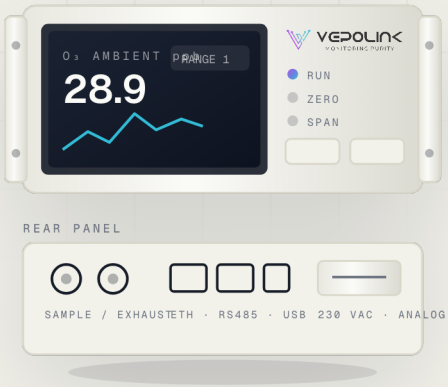
VEPOLINK
MONITORING PURITY

ENVIRONMENTAL INTELLIGENCE × GAS ANALYSIS

CONTINUOUS AMBIENT AIR QUALITY MONITORING

Ambient Ozone Gas Analyzer *UV Absorption*

Ground-level ozone measured down to parts per billion by UV absorption — with a high-efficiency scrubber for reference air, automatic zero correction and a year of logged data on board.



0.1 · 0.5 · 1 ppm
USER-SELECTABLE RANGES

7 inch
TFT LCD TOUCHSCREEN

1 year
ON-BOARD DATA LOGGING

UV ABSORPTION REFERENCE METHOD | **O₃ SCRUBBER** HIGH EFFICIENCY | **CPCB** ALIGNED REPORTING

0.5 ppb
LOWER DETECTABLE LIMIT

3
SELECTABLE MEASURING RANGES

60 s
TYPICAL RESPONSE TIME T90

32 bit
ARM PROCESSOR, LINUX BASED

01 OVERVIEW

Ground-level ozone, *read continuously.*

A continuous ambient ozone analyzer built on UV absorption — the accepted technique for regulatory O_3 measurement. A high-efficiency scrubber generates an ozone-free reference stream, and a specially coated specimen tube resists adhesion, giving a stable reading at the trace concentrations found at ground level.

A 32-bit ARM processor runs on Linux with a 7-inch touchscreen, automatic zero correction, and Ethernet, RS-232, RS-485, USB, DIO and analog interfaces — reporting to a station, SCADA and a regulatory server at once.

REPORTED CHANNELS

- O_3 INSTANT
- 1-MIN AVERAGE
- HOURLY AVERAGE
- 8-HR AVERAGE
- SAMPLE FLOW
- CELL TEMPERATURE
- UV LAMP INTENSITY
- SCRUBBER STATUS
- ZERO & SPAN STATUS
- FAULT DIAGNOSTICS

· PRODUCT FEATURES

Stable UV Source
A UV lamp selected for low sensitivity drift keeps span stable between calibrations and extends service intervals.

High-Efficiency O_3 Scrubber
An efficient scrubber strips ozone from a reference stream, giving a true zero baseline for a self-referencing UV measurement.

Touchscreen Operation
A 7-inch TFT LCD touch interface puts live values, trends, ranges and diagnostics in front of the operator.

One Year of Data On Board
An internal NAND-flash logger holds up to a year of one-minute averages, downloadable over USB or network.

02 MEASUREMENT PRINCIPLE

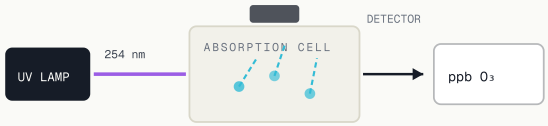
How UV absorption *counts O_3.*

Ozone absorbs ultraviolet light strongly at 254 nm. Comparing the light transmitted through the sample against a scrubbed, ozone-free reference gives a signal proportional to concentration — a direct optical measurement with no reagents in the sample path.

UV ABSORPTION

Absorb and Compare

UV LAMP · ABSORPTION CELL · DETECTOR



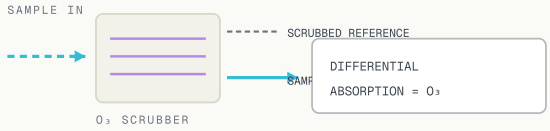
Sample air passes a cell illuminated at 254 nm. Ozone absorbs a fraction of that light proportional to its concentration; a detector measures the transmitted intensity.

`Absorption ∝ O₃ concentration`

REFERENCE

Scrubbed Baseline

HIGH-EFFICIENCY O₃ SCRUBBER



A high-efficiency scrubber strips ozone from a parallel air stream to create a true zero reference. The differential between sample and scrubbed reference isolates the ozone signal.

`Sample − Scrubbed = O₃`

STABILITY

Auto Zero & Compensation

AUTOMATIC BASELINE · T AND P CORRECTION



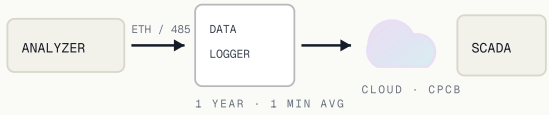
An automatic zero correction cycle re-establishes the baseline without operator intervention, while temperature and pressure compensation holds the reading steady as ambient conditions move through the day.

`Unattended long-term operation`

DATA PATH

Logging & Reporting

ETHERNET · RS-232 / 485 · USB · ANALOG



Readings are stored internally as one-minute averages for up to a year and transmitted live to a station data logger, plant SCADA or a pollution control board server.

`Download over USB or network`

03 SAMPLE PATH

From inlet to *reading.*

Ozone is reactive and short-lived on surfaces, so the sample path is built to move it to the detector fast and cleanly.

STEP 01

Filtered intake

Ambient air is drawn through a particulate filter and inert sample line into the analyzer at a controlled flow.

STEP 02

Coated specimen tube

A specially coated tube resists ozone adhesion, keeping the molecule intact between the inlet and the absorption cell.

STEP 03

Scrubbed reference

A parallel stream through the O_3 scrubber gives a true zero baseline against which the sample is compared.

STEP 04

Detect and report

UV absorption is measured, corrected for temperature and pressure, then displayed, logged and transmitted.

WHY THE COATING MATTERS

Ozone decomposes readily on untreated surfaces, losing concentration between the inlet and the cell. The specially coated specimen tube removes that loss path, which is what makes single-digit ppb measurement repeatable rather than theoretical.

RANGE SELECTION

Three ranges — 0–0.1, 0–0.5 and 0–1 ppm — are user selectable from the touchscreen. Background and urban sites usually run on the lowest range for resolution; summer photochemical episodes call for a wider range so peaks are captured without saturating.

WHY OZONE IS MONITORED

Ground-level ozone is not emitted directly but forms photochemically from NOx and volatile organic compounds under sunlight. It irritates the respiratory tract, reduces lung function on exposure and damages crops and vegetation at elevated levels. Continuous measurement is what lets a city track photochemical smog episodes and act on the precursors that drive them.

04 SPECIFICATIONS

Technical *specifications.*

MEASUREMENT

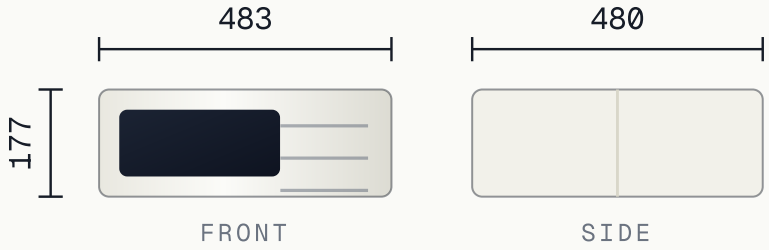
Parameter	Ambient ozone, O_3
Working Principle	UV absorption
Measuring Ranges	0 - 0.1 / 0.5 / 1 ppm, user selectable
Lower Detectable Limit	0.5 ppb
Precision	±1 % of reading
Linearity	±1 % of full scale
Zero Drift	< ±1 ppb per 24 hours
Span Drift	< ±1 % FS per 7 days
Response Time	T90 typically 60 s
Reference Generation	Built-in high-efficiency O_3 scrubber
Sample Path	Specially coated tube to prevent adhesion

SYSTEM · COMMS · BUILD

Display	7-inch TFT LCD with touch interface
Controller	32-bit ARM processor, Linux based multiprocessing
Zero Correction	Automatic · internal zero optional
Compensation	Temperature and pressure
Data Logger	NAND flash · up to 1 year of 1-minute averages
Interfaces	Ethernet · RS-232 · RS-485 · USB · DIO · analog out
Data Download	USB export or network transfer
Power Supply	230 VAC · 50 Hz
Mounting	19-inch rack or bench, station cabinet
Operating Conditions	5 - 40 °C, non-condensing
Enclosure	Weather-protected station cabinet for outdoor sites

INSTRUMENT DIMENSIONS

INDICATIVE FRONT & SIDE ELEVATION · MM



05 INSTALLATION & UPKEEP

Racked, calibrated, *left running.*

> The analyzer is installed in a station cabinet with the sample manifold, calibration source and data logger. Per CPCB practice, the sample inlet sits in free-flowing ambient air **3 to 10 m above ground**, with an inert, short sample line and a scheduled **zero and span check** traceable to certified gas.

01 · MOUNTING

19-inch rack or bench

Standard rack fitment alongside other CAAQMS analyzers, sharing the station's manifold, power and data logger.

02 · SAMPLE LINE

Inert and short

PTFE line with a particulate filter, kept short so ozone reaches the cell before it decomposes.

03 · ZERO AIR

Clean baseline source

A zero air unit supplies scrubbed air for the automatic zero cycle and for periodic manual verification.

04 · CALIBRATION

Multipoint traceable

Span with certified O_3 from a UV photometric calibrator or transfer standard, on the site's audit schedule.

05 · CONSUMABLES

Filter and scrubber

Routine upkeep is the inlet filter, periodic leak check and scrubber media inspection. No reagents and no wet chemistry.

06 · DIAGNOSTICS

Reported with the data

Lamp intensity, cell temperature and pressure, flow and zero/span status travel alongside every reading.

PART OF A FULL STATION

The ozone analyzer is normally specified with SO_2, NOx and CO analyzers, a multipoint calibrator and zero air unit, a particulate monitor and a meteorological mast. Sharing a time base, cabinet and data path, they report as one station record — which is what a pollution control board expects to receive.

COMMISSIONING & AUDIT

Commissioning covers leak and flow verification, a multipoint linearity check across the working range, and a documented zero and span baseline that becomes the reference for subsequent audits. Thereafter the automatic zero cycle carries day-to-day stability and the scheduled span check confirms it against certified gas, so the record stays defensible without a technician on site.

06 APPLICATIONS

Where it *works.*

Wherever sunlight and precursor gases combine, ozone has to be measured — for compliance, for public health reporting, and for tracking photochemical smog as it forms and moves.

Regulatory CAAQMS Networks	Urban & Traffic Corridors	Thermal Power Stations
Refineries & Petrochemicals	Chemical & Fertiliser Plants	Agricultural & Crop Studies
Ports & Marine Fuel Zones	EIA & Baseline Studies	Research & Air Quality Studies

WHAT THE DATA SUPPORTS

Continuous ozone records serve three audiences at once: regulators, who need hourly and 8-hour values against national standards; operations, who need to see how NOx and VOC emissions correlate with downwind ozone; and communities, who need advance warning on high-ozone days. One measurement, three conversations.

SITING FOR PHOTOCHEMICAL EPISODES

Because ozone forms downwind of its precursors, stations are usually placed to capture the plume some distance from the source, with a network of sites needed to track how a photochemical episode develops through the day. A wider measuring range keeps summer afternoon peaks on scale rather than clipped.

07 FAQ

Frequently asked *questions.*

What is the working principle?

UV photometric absorption. Ozone absorbs ultraviolet light at 254 nm in proportion to its concentration, measured against a scrubbed, ozone-free reference.

Which parameter does it measure?

Ambient ozone (O_3) concentration, for continuous ambient air quality monitoring in regulatory, industrial and research applications.

Is it suitable for continuous monitoring?

Yes. It is designed for real-time, continuous ozone monitoring in ambient and industrial environments, with automatic zero correction for unattended stability.

Does it meet CPCB requirements?

The analyzer is designed to meet CPCB and regulatory ambient air quality monitoring guidelines, and reports in the formats a state or central board server expects.

Can it be installed outdoors?

Yes, with a weather-resistant enclosure suitable for outdoor ambient air quality monitoring installations.

How does the scrubber improve accuracy?

It removes ozone from a parallel air stream to create a true zero reference, so the sample-versus-reference differential isolates the ozone signal from other UV-absorbing species.

What ranges are available?

0–0.1, 0–0.5 and 0–1 ppm, selectable by the user from the touchscreen to suit background, urban or photochemical-episode conditions.

How much data does it hold?

The internal NAND-flash logger stores up to one year of one-minute averaged data, exportable over USB or pulled across the network.

How does it integrate with a station?

Ethernet, RS-232, RS-485, DIO and analog outputs let the analyzer report to a station data logger, plant SCADA and a regulatory server simultaneously.

What maintenance does it need?

Inlet filter changes, periodic leak and flow checks, scrubber media inspection, and scheduled zero and span verification with certified gas.

This brochure is general product information. Configuration, dimensions and performance figures are indicative and confirmed at order. For installation guidelines and full technical detail, refer to the user manual supplied with the product. Specifications may change without notice as the product is improved.

VEPOLINK
MONITORING PURITY

GROUND-LEVEL OZONE, MEASURED CONTINUOUSLY

Know your O_3. *In real time.*

Tell us the site, the range you expect and who receives the data, and we'll configure the analyzer and the station around it — calibration, zero air, cabinet and reporting. Pick how you'd like to start.

01 **Talk to our engineers**
Speak directly to the team that specifies, installs and calibrates ambient gas analyzers. No call centre.
+91 99538 86178 →

02 **Request a station configuration**
RECOMMENDED
Send your site and reporting requirement — we'll spec ranges, calibration, zero air, cabinet and communications.
sales@vepolink.com →

03 **Book a demonstration**
See a running analyzer and the reporting dashboard with our engineering team.
Request a demo →

WWW.VEPOLINK.COM · SALES@VEPOLINK.COM